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[21MSCH3H04]

THIRD SEMESTER UNIVERSITY EXAMINATION, JANUARY - 2025

ORGANIC CHEMISTRY - III

Time: 03 HOURS **Maximum Marks:50**

		Blooms Level	CO	Marks
	SECTION – A			
1.	**Answer any FOUR of the following questions.**			**4×5=20**
a.	Examine the concept of functional group interconversion (FGI) and provide an example. Additionally, demonstrate one-group and two-group C-X disconnections with appropriate examples.	3	2	**05**
b.	Analyze the general reactivity of the Gilman reagent, providing an example.	4	3	**05**
c.	Write notes on the stereochemistry of electrocyclic reactions. Explain the terms conrotatory and disrotatory involved in it.	3	4	**05**
d.	Describe the mechanism involved in the following reaction.	4	5	**05**
e.	Explain with reactions Grignard reagents and hydrometallation.	2	6	**05**
	SECTION – B			
2.	**Answer any TWO of the following questions.**			**2×9=18**
a.	Write a note on-Classification and terminology of stereoselective synthesis.	3	1	**09**
b.	Examine the reactivity of 1,3-dithiane in terms of umpolung, including its mechanisms and applications in organic synthesis.	3	3	**09**
c.	**i.** Discuss the synthesis of organo-silicon compounds. **ii.** Explain the synthesis of organo-borane compounds.	4	6	**09**
	SECTION – C			
3.	**Answer the following questions.**			**1x12=12**
a.	**i.** Illustrate the retrosynthetic analysis and synthetic method for saccharin.	3	2	**04**
	ii. What is Claisen rearrangement? Write a note on its stereochemistry and olefin selectivity.	3	4	**04**
	iii. Explain two methods for the synthesis of benzimidazole.	2	5	**04**